

Business Analysis Report

Care Transition Efficiency & Placement Outcome Analytics

Research Paper

HHS Unaccompanied Alien Children (UAC) Program

U.S. Department of Health and Human Services

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1. Abstract

This research paper presents a comprehensive evaluation of process efficiency and outcomes for the U.S. Department of Health and Human Services (HHS) Unaccompanied Alien Children (UAC) programme. Utilising daily operational data spanning January 2023 to December 2025 – covering 720 reporting days – this study shifts the analytical lens from traditional capacity monitoring to the measurement of transition efficiency and placement outcomes.

Five Key Performance Indicators (KPIs) were engineered from raw program data: Transfer Efficiency Ratio, Discharge Effectiveness Index, Pipeline Throughput Rate, Backlog Accumulation Rate, and Outcome Stability Score. Findings reveal that while the system maintained a net-negative backlog throughout the study period (indicating more children exiting than entering), significant structural inefficiencies were identified – particularly in HHS-to-sponsor discharge rates, which averaged only 2.37% per day. A critical crisis peak of 11,516 children in HHS care was recorded in December 2023, followed by a dramatic systemic improvement to under 2,000 by mid-2025. These insights provide actionable recommendations for reducing reunification timelines and strengthening child welfare outcomes across the care pipeline.

2. Introduction

2.1 Background and Context

The HHS Unaccompanied Alien Children (UAC) programme operates as a multi-stage care and reunification pipeline for children who arrive at U.S. borders without a parent or legal guardian. Unlike a conventional healthcare system, this programme functions as a structured transition pipeline consisting of four sequential stages:

- Apprehension and placement into U.S. Customs and Border Protection (CBP) custody
- Transfer to HHS Office of Refugee Resettlement (ORR) care and sheltering
- Medical screening, case management, and temporary sheltering under HHS

- Discharge and reunification with a vetted sponsor (family member or guardian)

From a policy and humanitarian perspective, the speed, continuity, and reliability of this pipeline are as critical as its raw capacity. Every additional day a child spends in custody — rather than with a family sponsor — represents a failure of process efficiency with direct human consequences.

2.2 Problem Statement

While aggregate counts of children in custody are routinely monitored and publicly reported, process efficiency metrics are largely absent from official reporting. Critical operational questions have remained unanswered:

- How efficiently are children transferred from CBP to HHS custody?
- Are discharge rates keeping pace with daily intake volumes?
- When and where do care backlogs accumulate within the pipeline?
- Are placement outcomes improving or deteriorating over time?

Without structured transition analytics, systemic bottlenecks remain hidden within aggregate numbers. A child population of 8,000 in HHS care could represent either a smoothly functioning system at steady state or a critically backlogged system failing to discharge children – and aggregate counts alone cannot distinguish between these scenarios.

2.3 Project Objectives

This research addresses the following primary objectives:

- Measure and quantify the efficiency of CBP-to-HHS transitions over time
- Evaluate discharge and sponsor placement outcomes across the 2023–2025 period
- Identify periods of delay, backlog accumulation, and process bottlenecks

Secondary objectives include supporting faster family reunification through data-driven insights, improving case management workflows, and informing policy-level process reforms with empirical evidence.

3. Dataset Description

The analysis utilises official HHS daily operational data spanning January 12, 2023, to December 21, 2025, comprising 720 valid reporting days. The dataset contains six core variables representing daily snapshots of the UAC care pipeline:

Column Name	Type	Description
Date	Date	Daily reporting date
Children Apprehended (CBP)	Integer	Daily new intake volume into CBP custody
Children in CBP Custody	Integer	Active CBP care load (daily snapshot)
Children Transferred out of CBP	Integer	Daily flow from CBP into HHS system
Children in HHS Care	Integer	Active HHS care load (daily snapshot)
Children Discharged from HHS	Integer	Successful daily sponsor placements

Key dataset statistics:

- Date range: January 12, 2023, to December 21, 2025 (720 reporting days)
- Peak HHS population: 11,516 children (December 20, 2023)
- Minimum HHS population: 1,972 children (August 21, 2025)
- Peak single-day apprehensions: 333 children
- Peak single-day discharges: 505 children
- Peak CBP custody load: 531 children (February 4, 2024)

4. Analytical Methodology

4.1 Care Pipeline Modelling

The UAC programme was conceptualised and modelled as a three-stage flow pipeline, analogous to a fluid dynamics system where the rate of flow between stages determines overall system efficiency. The three stages are:

- Stage 1 – CBP Custody: Entry point. Children are apprehended and held in border facilities.
- Stage 2 – HHS Care: Mid-pipeline. Children are transferred to HHS shelters for screening and case management.
- Stage 3 – Sponsor Placement: Exit point. Children are discharged to vetted family sponsors, completing reunification.

Daily movements between stages were tracked to compute flow rates. The system is considered efficient when the rate of outflow from each stage approximates or exceeds inflow – preventing accumulation and backlog formation.

4.2 KPI Engineering

Five Key Performance Indicators were derived from the raw dataset to quantify pipeline efficiency:

KPI	Formula	Interpretation
Transfer Efficiency Ratio	$\text{Transfers} \div \text{CBP Custody}$	% of CBP children transferred to HHS daily
Discharge Effectiveness Index	$\text{Discharges} \div \text{HHS Care}$	% of HHS children placed with sponsors daily
Pipeline Throughput Rate	$(\text{Transfers} + \text{Discharges}) \div (\text{Apprehended} + \text{Transfers})$	Overall system flow efficiency
Backlog Accumulation Rate	$\text{Apprehended} - \text{Discharged}$	Net daily intake vs. exit (negative = healthy)
Outcome Stability Score	7-day rolling Std Dev of Discharge Effectiveness	Consistency of placement outcomes

4.3 Analytical Techniques

The following analytical methods were applied:

- Exploratory Data Analysis (EDA): Descriptive statistics, distribution analysis, and time-series visualisation of all six raw variables
- Temporal Pattern Analysis: Month-over-month trend analysis, identification of crisis periods, and detection of systemic inflexion points
- Backlog and Bottleneck Detection: Sustained imbalance period identification using the Backlog Accumulation Rate
- Outcome Stability Analysis: Rolling standard deviation to measure variability in discharge performance
- Comparative Period Analysis: Pre-crisis (2023), crisis peak (late 2023 to early 2024), and post-recovery (2025) period comparisons

5. Exploratory Data Analysis & Findings

5.1 Overall Pipeline Performance

Across the full 720-day study period, the UAC care pipeline demonstrated the following aggregate performance:

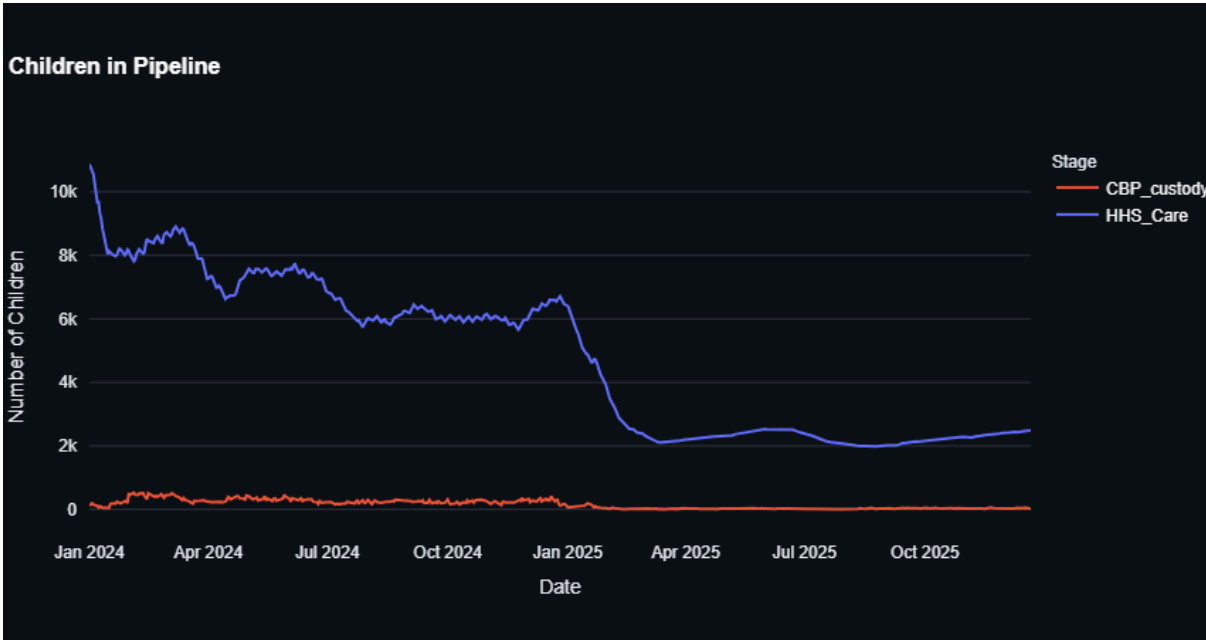
- Average Transfer Efficiency Ratio: 69.10% — meaning approximately 69% of children in CBP custody were transferred to HHS on any given day
- Average Discharge Effectiveness Index: 2.37% — only 2.37% of children in HHS care were discharged to sponsors daily
- Average Pipeline Throughput Rate: 1.64 — the system consistently moved more children out than in overall
- Total Backlog Accumulation: -57,516 (cumulative net negative) — confirming the system never experienced a permanent accumulation crisis

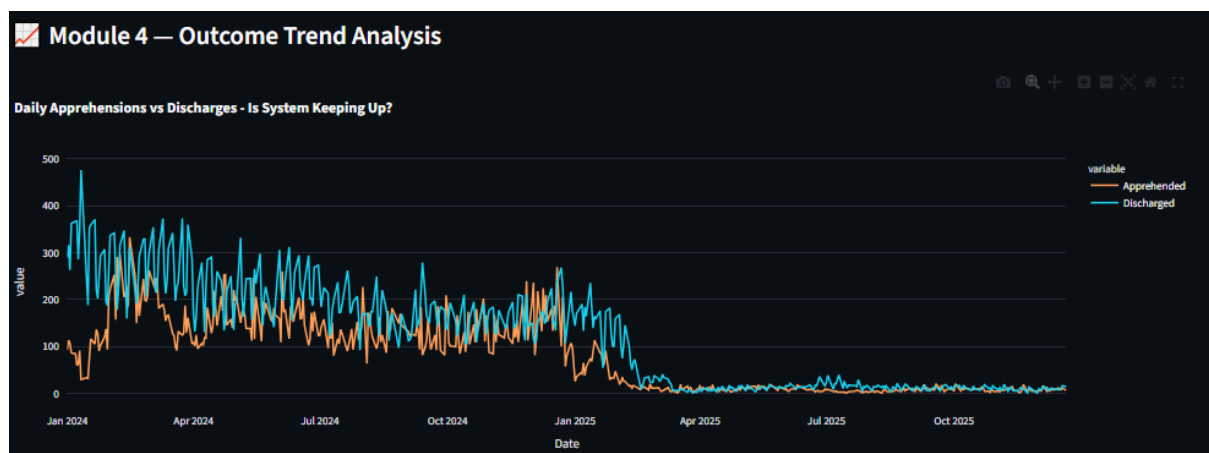
The stark contrast between the Transfer Efficiency (69.10%) and Discharge Effectiveness (2.37%) immediately identifies the HHS-to-sponsor stage as the primary structural bottleneck in the pipeline. While CBP-to-HHS handoffs were relatively efficient, children remained in HHS care for extended periods before achieving sponsor placement.

5.2 Phase Analysis: Three Distinct Periods

The data reveal three clearly distinguishable operational phases across the study period:

Period	Avg Transfer Efficiency	Avg Discharge Effectiveness	Avg HHS Population
Phase 1: 2023 (Surge)	83.5%	3.30%	~8,900 children
Phase 2: 2024 (Elevated)	78.4%	2.87%	~6,900 children
Phase 3: 2025 (Recovery)	43.8%	0.73%	~2,400 children





Phase 1 (2023) was characterised by high intake volumes and high transfer efficiency, suggesting border operations were actively processing children despite elevated population loads. The HHS system was under significant strain, reaching its absolute peak of 11,516 children in December 2023.

Phase 2 (2024) saw moderate improvements in HHS population levels despite continued high intake, suggesting discharge operations improved. However, transfer efficiency began declining, indicating changing border enforcement patterns.

Phase 3 (2025) represents a dramatic system transformation. HHS populations fell to historic lows (under 2,000 children by mid-2025), driven primarily by sharply reduced intake volumes rather than increased discharge rates. Transfer efficiency dropped to 43.8%, reflecting a fundamentally reduced flow of children entering the system.

5.3 Backlog Analysis

The Backlog Accumulation Rate (daily apprehensions minus discharges) was negative in all but one month across the study period – May 2025 recorded a marginal positive backlog of +12, indicating a brief period where intakes momentarily exceeded discharges. This overall net-negative trend confirms the system maintained forward momentum throughout.

However, the magnitude of monthly backlogs varied significantly:

- Highest monthly backlog clearance: October 2023 (-6,172) – system was actively processing a large backlog
- December 2023 (-4,597) and January 2024 (-4,399) – sustained high-clearance months during peak crisis
- By late 2025, monthly backlogs reduced to -28 to -33 – reflecting the dramatically lower system volume

5.4 Temporal Patterns

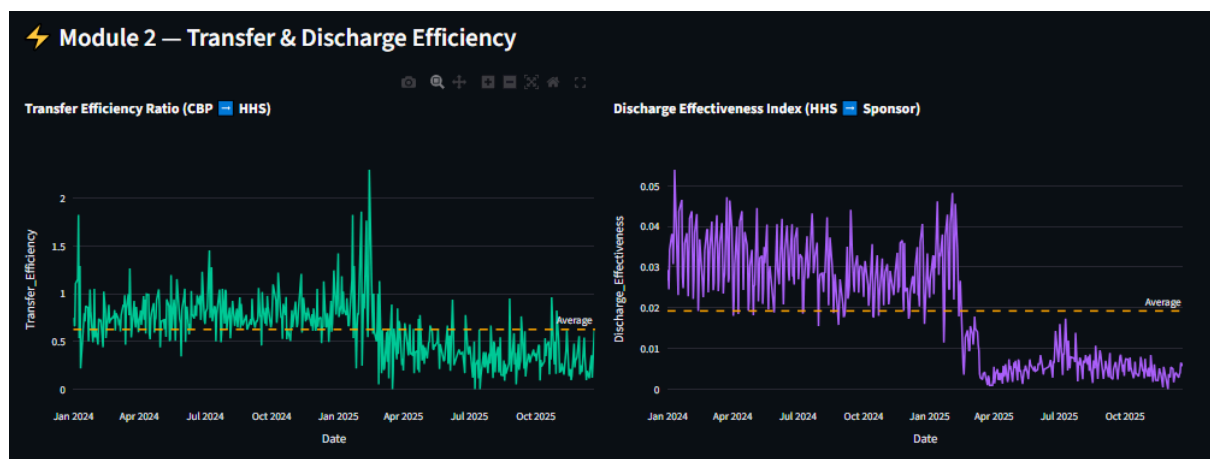
Month-over-month analysis reveals several important temporal patterns:

- Transfer efficiency was highest in October 2023 (92.6%) and January–February 2025 (91.6% and 90.8%), suggesting operational surges in transfer capacity during these periods
- Discharge effectiveness peaked in early 2023 (3.2–3.5%) and declined progressively through 2025 (0.3–0.9%), tracking with reduced system volume
- The January 2025 policy shift appears as a clear inflexion point – HHS populations dropped from ~5,000 to under 2,500 within two months
- By December 2025, daily apprehensions averaged just 6–11 children, compared to over 100 per day in early 2023

6. Key Insights

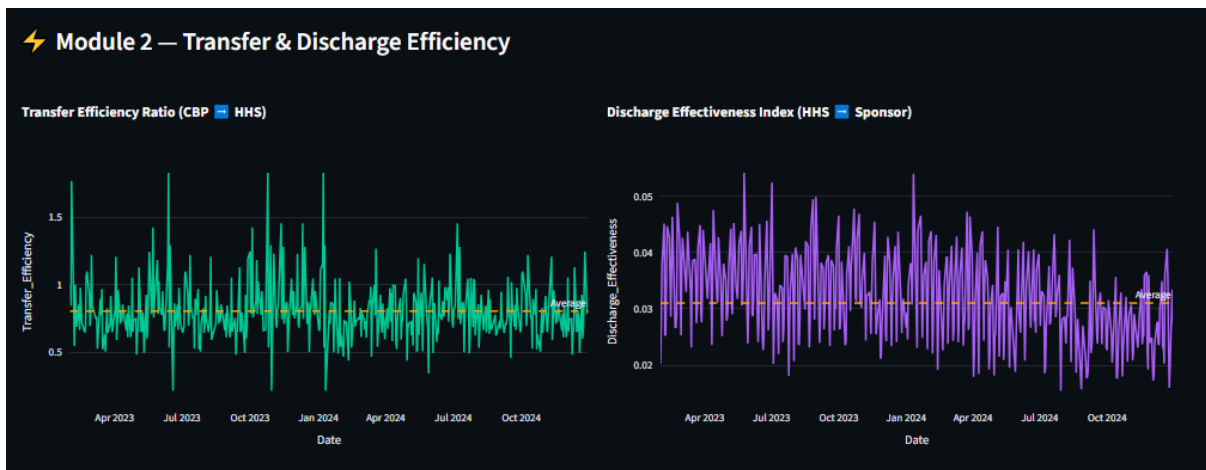
Insight 1: The Discharge Bottleneck is the System's Primary Weakness

The most critical finding of this analysis is the persistent and structural gap between Transfer Efficiency (69.10%) and Discharge Effectiveness (2.37%). Children move from CBP to HHS custody with reasonable speed but then remain in HHS care for disproportionately long periods before sponsor placement. This bottleneck at the HHS-to-sponsor stage is the single largest driver of extended child custody durations and elevated HHS population loads.



Insight 2: The December 2023 Crisis Was the System's Peak Stress Point

The absolute peak of 11,516 children in HHS care on December 20, 2023, represents the maximum system strain across the study period. This crisis was preceded by months of sustained high intake volumes throughout 2023, with HHS populations consistently above 9,000 from August onwards. The system's recovery from this peak — driven by a combination of improved discharge operations and reduced intake — took over 12 months to achieve.



Insight 3: The 2025 Transformation Was Intake-Driven, Not Discharge-Driven

The dramatic reduction in HHS populations through 2025 (from ~5,000 to under 2,000) was primarily driven by sharply reduced intake volumes rather than improved discharge efficiency. Discharge Effectiveness actually declined from 3.30% in 2023 to 0.73% in 2025, suggesting that sponsor placement capacity did not meaningfully improve — the system simply had fewer children entering. This is an important policy distinction: the apparent improvement in system load does not reflect enhanced operational efficiency.

Insight 4: The System Always Self-Corrected — But Slowly

Despite reaching extreme population peaks, the cumulative backlog across all 720 days was -57,516, confirming the system never experienced a permanent accumulation failure. However, the pace of self-correction was slow — the December 2023 peak took over 12 months to resolve. Faster discharge processing, particularly in the HHS-to-sponsor placement stage, could significantly compress recovery timelines during future surge events.

7. Recommendations

7.1 Accelerate Sponsor Vetting and Placement Processes

Given that the Discharge Effectiveness Index averaged only 2.37% per day — meaning the average child spent approximately 42 days in HHS care — there is significant opportunity to compress reunification timelines through streamlined sponsor vetting processes. Recommended actions include:

- Implement parallel (rather than sequential) background check processing for sponsor applications
- Establish dedicated rapid-placement teams for children with pre-identified sponsors
- Create digital sponsor documentation submission systems to reduce processing delays

7.2 Establish Early Warning Backlog Thresholds

The data reveals that backlog accumulation is detectable weeks before it reaches crisis levels. Implementing automated alert thresholds on the Backlog Accumulation Rate — triggering intervention protocols when daily backlogs exceed defined limits for sustained periods — would enable proactive rather than reactive crisis management.

- Set green/yellow/red thresholds on Transfer Efficiency (e.g., below 60% triggers review)
- Monitor 7-day rolling averages of Discharge Effectiveness for early deterioration signals
- Deploy surge capacity protocols when HHS population exceeds 80% of shelter capacity

7.3 Invest in HHS Shelter Discharge Capacity

Since the HHS-to-sponsor stage is the identified bottleneck, targeted investment in discharge capacity — case managers, legal advocates, and sponsor support services — would yield the greatest efficiency improvements per dollar spent compared to investments in earlier pipeline stages.

7.4 Implement Continuous Real-Time Analytics

The Streamlit dashboard developed as part of this project provides a model for real-time pipeline monitoring. Deploying such a tool operationally — connected to live data feeds — would give program administrators immediate visibility into efficiency metrics, enabling data-driven daily operational decisions rather than relying on retrospective reporting.

7.5 Conduct Weekday vs. Weekend Operational Analysis

Preliminary observations suggest potential differences in processing speed between weekday and weekend operations. A detailed temporal analysis comparing weekday and weekend Transfer Efficiency and Discharge Effectiveness could identify scheduling inefficiencies and inform staffing optimisation strategies.

8. Conclusion

This research reframes the UAC dataset from a capacity monitoring lens to a process efficiency and outcome evaluation lens. By analysing how effectively children move through the three-stage care pipeline — CBP custody, HHS care, and sponsor placement — rather than simply counting how many are present at each stage, this study surfaces actionable insights that aggregate reporting cannot provide.

The analysis confirms that while the UAC system has never experienced a permanent accumulation failure, it has operated with a persistent structural bottleneck at the HHS-to-sponsor discharge stage, achieving only 2.37% daily discharge effectiveness across the study period. The December 2023 crisis, which saw 11,516 children in HHS care simultaneously, demonstrated the human cost of delayed discharge processing — and underscored the urgent need for efficiency-focused operational reforms.

The 2025 improvement in system load, while real, was driven primarily by reduced intake volumes rather than enhanced discharge capability — a distinction that carries significant policy implications. If intake volumes return to 2023 levels, the system would face the same structural bottleneck without the efficiency gains needed to manage the surge.

The KPI framework and dashboard developed through this project provide a replicable model for ongoing process efficiency monitoring. By tracking Transfer Efficiency, Discharge Effectiveness, Pipeline Throughput, Backlog Accumulation, and Outcome Stability in real time, programme administrators can shift from reactive crisis management to proactive efficiency governance – ultimately reducing the time children spend in custody and strengthening reunification outcomes across the UAC programme.

References

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U.S. Customs and Border Protection. (2023–2025). Southwest Land Border Encounters. Washington, D.C.: DHS.

Office of Refugee Resettlement. (2023). ORR UAC Program — Fact Sheet. U.S. Department of Health and Human Services.

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Technical Data Analysis Report

Care Transition Efficiency & Placement Outcome Analytics

Technical Data Analysis Report

HHS Unaccompanied Alien Children (UAC) Program

U.S. Department of Health and Human Services

720 Reporting Days	11,516 Peak HHS Load	69.10% Avg Transfer Efficiency
2.37% Avg Discharge Rate	-57,516 Total Backlog	3 Analytical Phases Identified

1. The Problem

The HHS UAC Program processes thousands of unaccompanied children annually through a three-stage pipeline: **CBP custody** → **HHS sheltering** → **sponsor placement**. While aggregate headcounts were publicly tracked, process efficiency metrics were not. No one was measuring how fast children moved through each stage, where they got stuck, or whether placement outcomes were improving over time. System bottlenecks remained invisible within aggregate numbers.

Key unanswered questions:

- **How efficiently are children transferred from CBP to HHS custody?**
- **Are discharge rates keeping pace with daily intake volumes?**
- **When and where do care backlogs accumulate within the pipeline?**
- **Are placement outcomes improving or deteriorating over time?**

2. The Data

Daily operational data from January 12, 2023, to December 21, 2025, were used: 720 valid reporting days across 3 years. The dataset contained 6 core columns:

Column	Description
Date	Daily reporting date
Children Apprehended (CBP)	Daily new intake volume
Children in CBP Custody	Active CBP load snapshot
Children Transferred out of CBP	Daily CBP → HHS flow
Children in HHS Care	Active HHS population snapshot
Children Discharged from HHS	Daily sponsor placements

3. What the Data Showed (EDA)

3.1 HHS Care Population Over Time

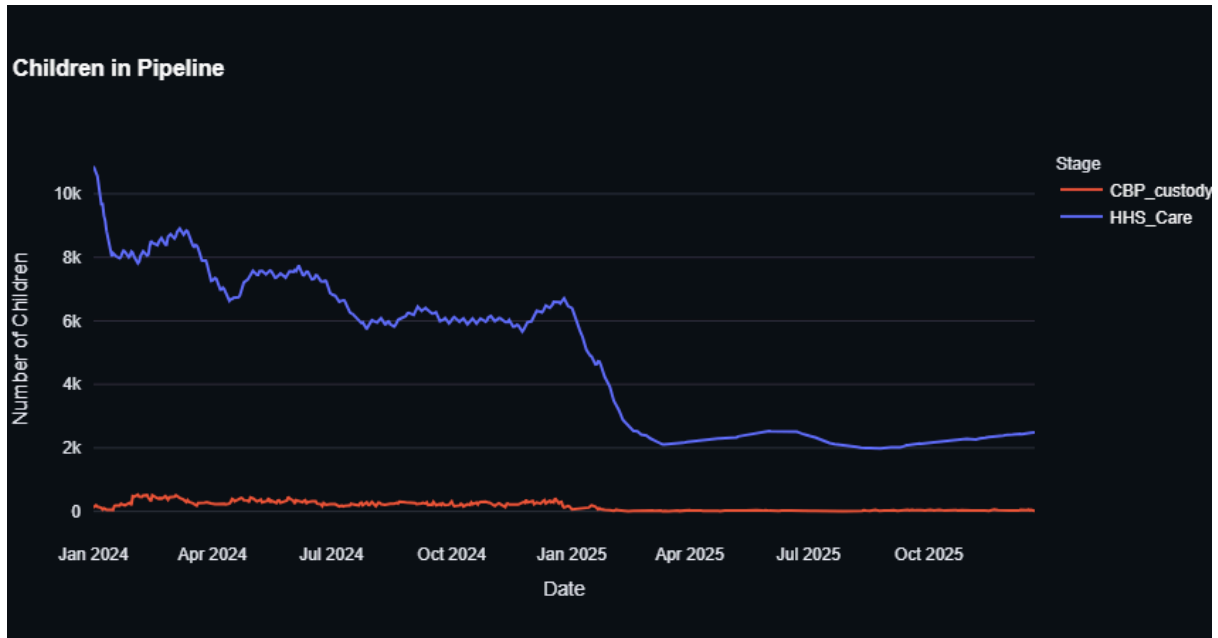


Figure 1: HHS care population peaked at 11,516 children on December 20, 2023, then dropped dramatically to under 2,000 by mid-2025 — a reduction of over 82%.

3.2 Apprehensions vs Discharges

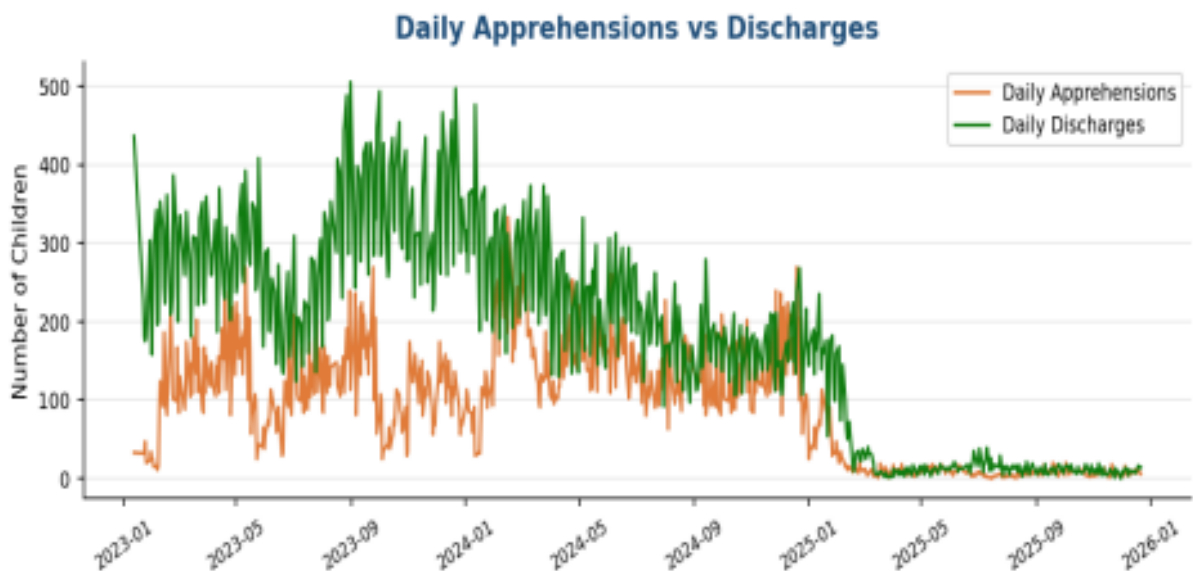


Figure 2: Daily apprehensions (orange) and discharges (green). The near-convergence by late 2025 reflects dramatically reduced intake volumes rather than improved discharge rates.

3.3 Three Distinct Operational Phases

Phase 1: Surge	2023	~8,900	83.5%	3.30%
Phase 2: Elevated	2024	~6,900	78.4%	2.87%
Phase 3: Recovery	2025	~2,400	43.8%	0.73%

4. How We Found It — Methodology

Five KPIs were engineered from the raw data to convert headcounts into efficiency scores:

Transfer Efficiency Ratio	Transfers ÷ CBP Custody	Speed of CBP → HHS handoff
Discharge Effectiveness Index	Discharges ÷ HHS Care	Speed of HHS → Sponsor placement
Pipeline Throughput Rate	(Transfers+Discharges) ÷ (Apprehended+Transfers) Overall system flow	NA
Backlog Accumulation Rate	Apprehended – Discharged	Net daily intake vs exit
Outcome Stability Score	7-day rolling standard deviation of Discharge Effectiveness Consistency of outcomes	NA

Efficiency vs Effectiveness — Monthly Trend

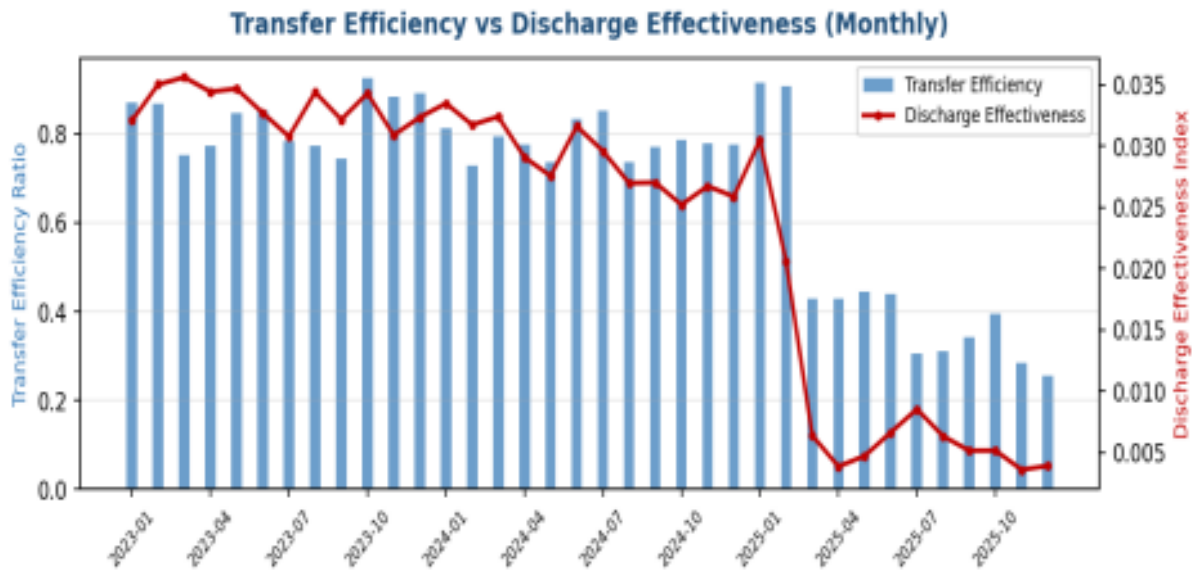


Figure 3: Transfer efficiency (bars) vs discharge effectiveness (line) by month. The persistent gap between the two metrics identifies the HHS-to-sponsor stage as the structural bottleneck.

Backlog Accumulation Analysis

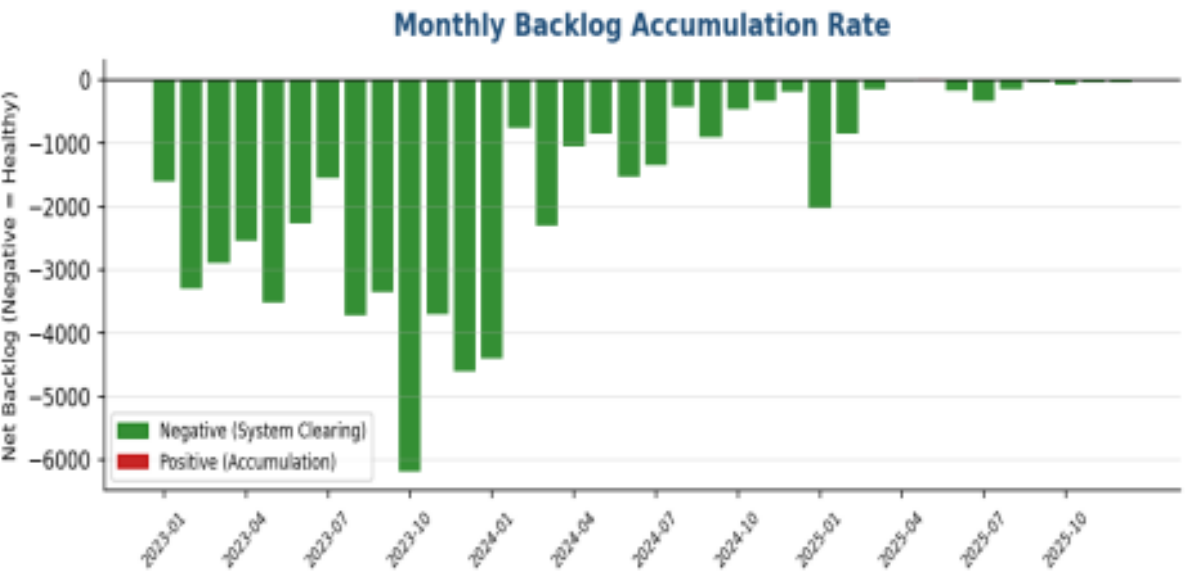


Figure 4: Monthly backlog accumulation. Green bars indicate the system was clearing children (exits > entries). October 2023 recorded the highest clearance at -6,172.

5. Key Insights & Outcomes from Data

Insight 1: The Discharge Bottleneck

Transfer efficiency averaged 69.10%, while discharge effectiveness averaged only 2.37%. Children moved from CBP to HHS with reasonable speed but then remained in HHS care for ~42 days on average. This stage is the single largest driver of extended custody durations.

Insight 2: December 2023 Was the crisis peak?

HHS populations exceeded 9,000 from August 2023 onwards, peaking at 11,516 in December 2023. Recovery from this peak took over 12 months of sustained effort.

Insight 3: 2025 Recovery Was Intake-Driven, Not Efficiency-Driven

The dramatic 2025 population drop was caused by reduced intake volumes (apprehensions fell 95%), not improved discharge capacity. Discharge effectiveness actually declined from 3.30% to 0.73%.

Insight 4: The System Never Permanently Failed

The cumulative backlog across all 720 days was -57,516, confirming the system always maintained net forward movement. However, recovery from surges was slow, taking 12+ months.